

What Can Cooperative Agents Forget?

Garbling, Stable Decisions, and Source-Bound Program Games

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Cooperating agents need information about their partners, but need not retain every distinction in that information. We study finite program games with authentic committed sources and explicit joint observation laws. We characterize compression that preserves simultaneous casewise guarantees: statically, merged receipts must admit common optimal decisions; adaptively, a retained policy must avoid optimal-action conflicts at the histories it actually reaches. Robust regret constraints provide quantitative compression certificates. A repeated heterogeneous construction preserves cooperative guarantees while merging different source posteriors, whereas feedback can make identity indispensable. Equal expected payoffs and pairwise temporal laws can also conceal different downside risks. Experiments with Qwen3-written programs distinguish these observables: fitness selection favors low coordination, reduced temptation increases high cooperation, and a risk-cap instruction improves individual compliance without protecting every population member. The results connect decision-relevant compression to source-level incentives while separating co-operation, equilibrium, and protection against replacement opponents.

1. Introduction

Before cooperating, a software agent may inspect its partner’s source, receive a behavioral signal, or exchange readiness messages. Full disclosure distinguishes many implementations that would support the same interaction. Compressing those observations can reduce memory or communication, but can also erase distinctions required for safe or incentive-compatible behavior. We ask: *what can an agent forget while preserving the guarantees of cooperation?*

The preservation target matters. Matching expected payoffs need not preserve downside probabilities. A Nash equilibrium can be vulnerable to a harmful replacement opponent whose behavior is not a profitable deviation. Reproducing a posterior is sufficient for many decisions, but is stronger than preserving a specified collection of cooperative guarantees. We therefore describe performance by a vector of casewise monitors: cooperation, normalized payoff, and absence of downside against each admitted opponent and environment. One policy must satisfy all coordinates simultaneously.

Our analysis has three components. First, we give exact and robust conflict conditions for merging finite observations. The smallest unconstrained alphabet is determined by incompatible collections of receipts; randomized decisions can enlarge the robustness radius even when stochastic garbling cannot reduce the alphabet. Second, we characterize adaptive preservation through occupied Bellman regret and construct a repeated heterogeneous class that loses posterior information without losing its useful guarantee region. Third, we evaluate model-written programs in a finite non-prisoner’s-dilemma game. The populations exhibit distinct responses in cooperation, tested deviation gain, and replacement risk, including average risk improvements that leave the worst-writer guarantee unchanged.

Programs are both strategies and observable objects. Every source deviation is evaluated using its authentic new quote and its actual environmental kernel. Optimizing commands at a fixed control port is consequently distinct from replacing the source exposing that port. This distinction is maintained throughout the theory and experiments.

2. Source-bound games and useful guarantees

2.1. The game and its information interface

A finite stage game has physical actions A_1, A_2 and payoffs $u_i : A_1 \times A_2 \rightarrow \mathbb{R}$. Each player commits an immutable source string s_i , interpreted by a trusted terminating executor. At profile $s = (s_1, s_2)$, a joint causal kernel K_s specifies authentic quotations, message delivery, private or shared randomness, and persistent environmental state. Together with the programs it induces a trace law P_s . A unilateral source deviation d_i uses $P_{(d_i, s_{-i})}$, including the deviating source's quotation. For declared source families, Nash equilibrium means

$$\mathbb{E}_{P_s} u_i \geq \mathbb{E}_{P_{(d_i, s_{-i})}} u_i \quad \text{for every admitted } d_i \text{ and } i.$$

Our running game has secure opt-out O, low coordination L, and high cooperation H:

	O	L	H
O	(1, 1)	(1, 0)	(1, 0)
L	(0, 1)	(2, 2)	(6, 0)
H	(0, 1)	(0, 5)	(4, 3)

(1)

Opt-out guarantees own payoff 1; *downside* is the event $u_i < 1$. Low coordination is a strict stage equilibrium. High cooperation benefits both players, but either can exploit it by choosing L.

The basic protocol is commit sources, draw joint authentic quotes, send and deliver readiness bits, and select physical actions. Readiness cannot use future delivery, and neither physical action is revealed before both are chosen. A *checked gate* plays H exactly when its own acceptance and the delivered peer bit are one, and otherwise plays O. A sound certificate establishes the physical obligation of the actually executed source in a specified bounded language; an unverified readiness claim does not.

A garbler $\Gamma(z \mid e)$ replaces fine evidence e by retained evidence z , with no unobserved case input or visible random seed unless that seed is part of z . A fine agent can simulate the garbler. In a repeated interface, the retained history must be causally updated and retain its own commands and observations. Joint resource dependence is preserved rather than inferred from marginal probabilities.

2.2. The guarantee region

At a fixed, already committed control port, let $v_{c,j}(\pi) = \mathbb{E}_c^\pi Y_j$ for actual cases c and trace monitors $Y_j \in [0, 1]$. Larger coordinates are preferable: payoff is positively normalized and risk enters through its complement. The downward guarantee region is

$$\mathcal{D} = \{g \in [0, 1]^q : \exists \pi, g \leq v(\pi)\}.$$

Independent initial policy mixtures make the finite performance image convex. Fine and retained regions satisfy $\mathcal{D}_C \subseteq \mathcal{D}_F$. They agree exactly when their supporting values agree for every nonnegative weight $w \in \mathbb{R}_+^q$. This follows from convex separation: a point outside a compact downward region has a nonnegative separating normal. Thus preserving useful guarantees tests fewer directions than reproducing all performance vectors or all source posteriors.

3. Exact and robust observation compression

3.1. Common optimal decisions

Let $R(c, e) = \mathbb{P}(e \mid c)$ be a finite evidence matrix and let k decisions be available. A randomized table $T(e, a)$ gives decision law RT . For an objective $W(c, a)$, including any desired case weights, put

$$b_e(a) = \sum_c R(c, e)W(c, a), \quad V_R(W) = \max_T \sum_{c,a} (RT)(c, a)W(c, a).$$

Garbling permits only tables $T = \Gamma U$.

Theorem 3.1 (Exact conflicts and alphabet size). *The optimal score is $V_R(W) = \sum_e \max_a b_e(a)$, and its loss under garbling is*

$$\sum_z \left[\sum_e \Gamma(z \mid e) \max_a b_e(a) - \max_a \sum_e \Gamma(z \mid e) b_e(a) \right]. \quad (2)$$

Each term is nonnegative and vanishes exactly when the positive output support $B_z = \{e : \Gamma(z \mid e) > 0\}$ admits a common optimal action. For a declared objective family, form a hypergraph whose edges are minimal receipt sets lacking a common optimum for at least one objective. If all evidence maps are allowed without output-frequency constraints, the minimum lossless alphabet is its chromatic number. Stochastic garbling cannot reduce this number, and every minimal edge has at most k receipts.

For two separated continuation plans, performance forms the zonotope $b + \sum_e [0, v_e]$: choosing a plan at one receipt does not change another receipt's likelihood or continuation law. The retained image combines generators as $b + \sum_z [0, \sum_e \Gamma(z \mid e) v_e]$. In a useful direction $w \geq 0$, its exact support loss is

$$L_w = \sum_z \min \left\{ \sum_e \Gamma(z \mid e) [w^\top v_e]_+, \sum_e \Gamma(z \mid e) [-w^\top v_e]_+ \right\}. \quad (3)$$

Consequently the downward region survives exactly when no output support contains opposing signs in any nonnegative direction. The conflict hypergraph is then a graph. Nonproportional generators in the same preferred orthant may merge, although the full zonotope changes.

Example 3.2 (Posterior loss without guarantee loss). Two cooperative sources h_1, h_2 and two outsiders d_1, d_2 are authentically revealed with probability ρ and otherwise produce the opaque receipt u . The decision is activate or reject. Cooperative activation and outsider rejection coordinates have generators

$$\begin{aligned} b &= (0, 0, 1, 1), & v_u &= (1 - \rho)(1, 1, -1, -1), \\ v_{h_1} &= \rho(1, 0, 0, 0), & v_{h_2} &= \rho(0, 1, 0, 0), \\ v_{d_1} &= -\rho(0, 0, 1, 0), & v_{d_2} &= -\rho(0, 0, 0, 1). \end{aligned}$$

Merge the two cooperative receipts, merge the two outsider receipts, and retain opaque. Accepting the cooperative label, rejecting the outsider label, and preserving the original opaque acceptance probability dominates every fine rule. The whole four-coordinate guarantee region survives. Under an equal prior, a fine cooperative quote gives a point mass and the retained label gives posterior $(1/2, 1/2, 0, 0)$. Signed support in direction $(1, -1, 0, 0)$ decreases by ρ . The conflict graph is complete tripartite, so three retained symbols are necessary and sufficient. Common coverage α with outsider false acceptance at most δ has frontier $\alpha_{\max} = \min\{1, \rho + \delta\}$.

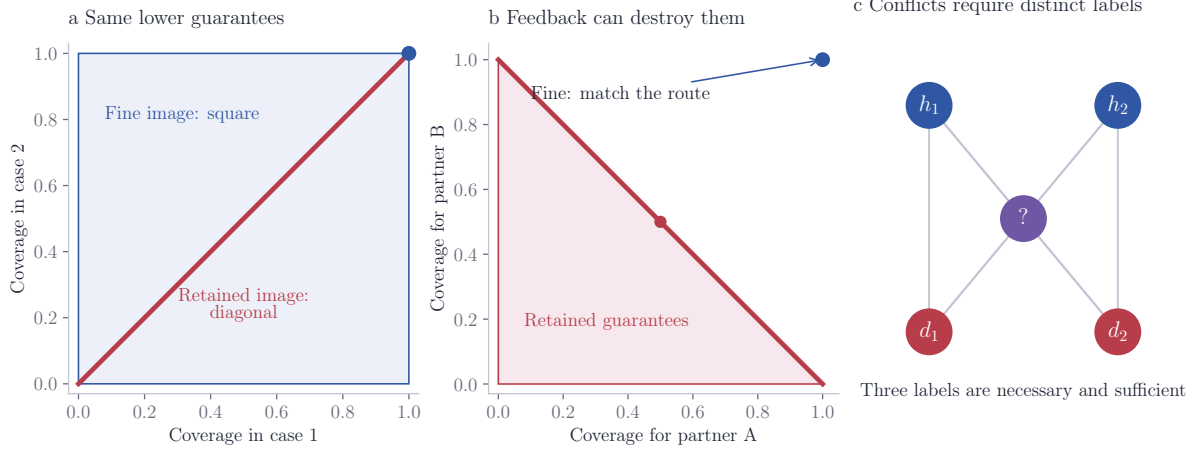


Figure 1: Decision-relevant geometry. **a**, merging two source-specific decisions changes a square to a diagonal while preserving its downward closure. **b**, route-dependent partners instead lose the upper cooperation corner after merging. **c**, the five receipts in Example 3.2 require three labels; edges mark opposing useful decisions.

3.2. Robust admissible decisions

Exact agreement can disappear under a small score perturbation. Suppose $\|\tilde{b}_e - b_e\|_\infty \leq r s_e$, with $s_e > 0$, and require one decoder $q \in \Delta_k$ to have regret at most η_e for every perturbation. Perturbations are not revealed before decoding.

Theorem 3.3 (Robust conflict cells). *The exact worst-case regret is*

$$\sup_{\|\tilde{b}_e - b_e\|_\infty \leq r s_e} \left(\max_a \tilde{b}_e(a) - \tilde{b}_e^\top q \right) = \max_a \{b_e(a) - b_e^\top q + 2r s_e(1 - q_a)\}.$$

Thus robust decoders form the polytope

$$C_e(r, \eta_e) = \{q \in \Delta_k : b_e(a) - b_e^\top q + 2r s_e(1 - q_a) \leq \eta_e \forall a\}. \quad (4)$$

A cell B is valid exactly when $\bigcap_{e \in B} C_e(r, \eta_e) \neq \emptyset$. Every minimal infeasible cell has at most k receipts. When no singleton is infeasible, the smallest branchwise-valid alphabet is the chromatic number of these conflict hyperedges, again without output quotas. Fixed-radius validity is a linear feasibility problem, monotone in r .

For a fixed deterministic action a and zero tolerance, the exact cell radius is

$$\min_{e \in B} \min_{b \neq a} \frac{b_e(a) - b_e(b)}{2s_e},$$

when nonnegative. Randomized decisions can do better at positive tolerance. For scores $(1, 0.8, 0)$ and $(0.8, 1, 0)$, scales one and tolerance 0.35, the decoder $(1/2, 1/2, 0)$ is valid up to radius 0.25, while the best deterministic decoder permits only 0.075. Adding the two preferred-action inequalities shows that no decoder exceeds 0.25.

The guarantees hold at each receipt and each possible output branch. Averaging regret over branches or imposing output quotas changes the problem. For example, receipts of masses $(1/4, 1/4, 1/2)$ with action rewards $(2, 0, 1)$ and $(0, 2, 1)$ must use two equally frequent labels. Splitting the third receipt equally achieves the fine optimum $3/2$; every deterministic balanced partition achieves only 1. The no-quota hypothesis is therefore essential.

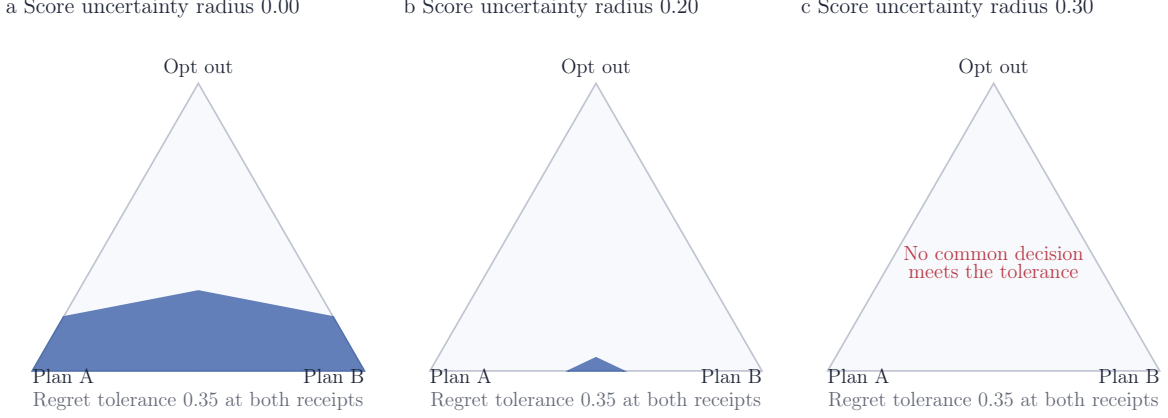


Figure 2: Common randomized decoders under increasing score uncertainty for the two-receipt example. Each triangle is the probability simplex over three decisions; shaded regions satisfy both receipts’ regret constraints. The common region contracts to $(1/2, 1/2, 0)$ at radius 0.25 and is empty above it.

4. Adaptive guarantees and repeated cooperation

4.1. Occupied continuation conflicts

Fix a finite controlled history tree of horizon T , actual source/case kernels, and every legal causal policy at the bound port. The environment is blind to private policy seeds and observation-mode selectors except through commands. Retained history $z = m_n(h)$ has coherent causal updates and perfect recall; merged fine histories have the same legal menu. Lifetime monitors include their persistent bad-event flags.

For casewise weights $w \geq 0$, write $J_w(\pi) = \sum_{c,j} w_{c,j} \mathbb{E}_c^\pi Y_j$. Choose a full-support reference case distribution ϖ and terminal scalar reward $\sum_j w_{c,j} Y_j / \varpi_c$. Its expectation is J_w ; the guarantees remain casewise. Let $V_w(h)$ and $Q_w(h, a)$ be fine optimal continuation values under the actual conditional laws, and let $d_{\bar{\pi}}(h)$ be fine-history occupancy under a lifted retained policy.

Theorem 4.1 (Adaptive preservation). *For every retained policy $\bar{\pi}$,*

$$\max_{\pi} J_w(\pi) - J_w(\bar{\pi}) = \sum_{n < T} \sum_{h,a} d_{\bar{\pi}}(h) \bar{\pi}(a \mid m_n(h)) [V_w(h) - Q_w(h, a)]. \quad (5)$$

The entire downward guarantee region is preserved if and only if, for every $w \geq 0$, one causal retained policy uses only fine-optimal commands at every fine history it reaches with positive probability. Equivalently, every deterministic fine-policy vector is dominated by a mixture of deterministic retained-policy vectors.

The criterion charges reached histories, not every conceivable conflicting cell. An earlier optimal safe command can make a later conflict irrelevant. The equivalent finite test seeks mixture weights $\lambda_m \geq 0$, $\sum_m \lambda_m = 1$, with $\sum_m \lambda_m u^m \geq v$ for each fine vector v and retained vectors u^m . Dynamic history compression also requires coherent updating; its policy image is generally not a zonotope.

An approximate version translates support loss into a simultaneous error budget. If h_F, h_C are fine and retained supporting values, every fine vector has a retained vector within ε from below in every coordinate exactly when

$$h_F(w) - h_C(w) \leq \varepsilon \|w\|_1 \quad \forall w \geq 0.$$

The least common error is $\max_{w \geq 0, \|w\|_1 \leq 1} [h_F(w) - h_C(w)]$. Receiptwise regret bounds sum in the static problem; occupied epochwise Bellman regret bounds sum in the adaptive problem.

Proposition 4.2 (Common dominating continuations). *Suppose fine and retained histories coincide until an exogenous first sound, remembered certificate. At every compatible certified prefix, its retained label selects one executable tail dominating every fine continuation in all specified conditional expected monitors, including existing monitor state. Then replacing any fine policy by that tail preserves its casewise guarantees, so $\mathcal{D}_F = \mathcal{D}_C$.*

The condition is stronger than physical safety. If partner A becomes ready only after route a and partner B only after route b , both can obey the same safe joint gate. Fine identity permits cooperation $(1, 1)$; one common label permits only $(t, 1 - t)$, with best uniform coverage $1/2$. The missing object is a common useful continuation.

4.2. An actual heterogeneous class

Consider cooperative sources with physical gate H iff both reliably delivered readiness bits are one, otherwise O. At actual profile s , disclosure opens jointly and persistently at time τ . Initial blocking has probability q_s , and each later blocked epoch opens with hazard r_s . Thus the undisclosed probability at epoch n is $q_s(1 - r_s)^n$. Before opening both modes receive opaque; afterward fine mode quotes the actual counterpart and coarse mode gives a sound structural class bit. Prepared mode selectors do not enter the resource kernel.

Cooperative sources reject a negative certificate. After a positive certificate, their readiness depends only on their own parameters, clock, and ticket from the actual joint tape, for example

$$b_{i,n} = \mathbf{1}\{(u_{i,n} + \ell_i + n) \bmod 4 < k_i\}, \quad u_{i,n} \in \{0, 1, 2, 3\}, \quad k_i \in \{0, \dots, 4\}.$$

Opaque readiness may instead use causal feedback. Outside sources may send arbitrary legal causal bits and choose any physical action after readiness, including delayed or history-conditioned betrayal. Every outside source receives a negative certificate. Source deviations recompute the authentic quotes and their profile-specific resource rows.

Theorem 4.3 (Mixed modes, safety, and source equilibria). *Changing cooperative receivers' prepared fine/coarse modes preserves their whole joint readiness, action, and payoff trace law, holding outside modes fixed. This includes simultaneous changes and causal selectors. Two cooperative sources have zero lifetime downside and weighted payoffs*

$$U_1 = 1 + 3 \sum_n d_n c_{s,n}, \quad U_2 = 1 + 2 \sum_n d_n c_{s,n},$$

where $d_n \geq 0$, $\sum_n d_n = 1$, and $c_{s,n}$ is actual joint activation. Against an outsider, cooperative lifetime downside is at most q_s and outsider role j earns at most

$$1 + (M_j - 1)\kappa_s(d), \quad \kappa_s(d) = q_s \sum_n d_n (1 - r_s)^n, \quad (M_1, M_2) = (6, 5). \quad (6)$$

Full-readiness cooperative pairs earn $(4, 3)$ and are source-commitment equilibria if every actual outside-deviated profile satisfies $5\kappa_s \leq 3$ for role 1 and $4\kappa_s \leq 2$ for role 2. Independent mixtures of full-readiness implementations also satisfy the equilibrium conditions when the bounds are uniform over their support.

A three-round instance has six heterogeneous cooperative sources and ten outsiders. Its actual complete-source checksum selects $(q_s, r_s) = (1/3, 1/3)$ or $(1/4, 1/2)$. Equal round weights give $\kappa_s \leq 19/81$, outsider ceilings $(176/81, 157/81)$, and outside-deviation margins $(148/81, 86/81)$ below full cooperation. Shared and independent tickets must remain distinct: a half-ready pair has joint cooperation $1/2$ under shared tickets and $1/4$ independently.

For the optimized guarantee region, bind a separate readiness-command port with the same physical gate and explicitly include it in partners' sound accepted class. After a positive certificate its arbitrary fine continuation is dominated by readiness one, because the autonomous

partner's certified schedule ignores the receiver's subsequent path. After a negative certificate, readiness zero prevents new exposure and downside. Proposition 4.2 therefore preserves cooperative coverage/payoff lower bounds and outsider exposure/safety guarantees at this port. Lower payoff guarantees against outsiders are not included. Under equal weights on two distinct cooperative sources with the fixed disclosure row, fine quotes yield point masses while coarse positive receipts yield posterior $(1/2, 1/2)$. Different certified readiness schedules may be merged as well. This full-policy-port claim is distinct from the autonomous catalog's preserved realized traces.

5. Incentives, degradation, and downside

5.1. Joint delivery and source-level margins

Let two checked gates jointly recognize with probability p . Conditional on recognition, suppose readiness delivery has equal marginals r and joint success κ . The physical law has probabilities $p\kappa$ on HH, $p(r - \kappa)$ on each mismatch, and $1 - p(2r - \kappa)$ on OO. Therefore

$$U_1 = 1 + p(4\kappa - r), \quad U_2 = 1 + p(3\kappa - r), \quad D_1 = D_2 = p(r - \kappa). \quad (7)$$

Shared delivery has $\kappa = r$ and no mismatch; independent delivery has $\kappa = r^2$ and downside $pr(1 - r)$. If recognition and delivery are dependent, these are conditional quantities rather than products of marginal rates.

At fixed authentic sources and quote law, changing only coupling from κ' to κ changes any outcome monitor g by

$$(\kappa - \kappa')\mathbb{E}_e[g(o_{11}) - g(o_{10}) - g(o_{01}) + g(o_{00})], \quad (8)$$

where o_{xy} is the outcome at quote atom e under delivery bits x, y . This follows directly from the four joint delivery probabilities and applies even to programs outside the checked class.

These bounds extend to a finite game with target h , fallback f , baseline $v_i = u_i(f)$, and nonnegative surplus $g_i = u_i(h) - v_i$. Checked pairs with activation c earn $v_i + g_i c$. A secure fallback means $u_i(f_i, a_{-i}) \geq v_i$ for all opponent actions. If outsider exposure is at most δ , downside is at most δ , and outsider payoff is at most

$$B_j^f + [B_j^h - B_j^f]_+ \delta, \quad B_j^f = \max_{a_j} u_j(a_j, f_{-j}), \quad B_j^h = \max_{a_j} u_j(a_j, h_{-j}).$$

The outsider can deliberately send readiness one, so exposure is not squared. Security of one's fallback and the outsider's best reply against it are different requirements. Within-class source deviations still need their own actual-profile comparisons.

A replacement certified at every joint controlled row, including persistent state and all deviated profiles, also supports quantitative composition. Initial and successive mismatch bounds $\epsilon_0, \dots, \epsilon_T$ imply trace total variation at most $\delta_T = 1 - \prod_{n=0}^T (1 - \epsilon_n) \leq \sum_n \epsilon_n$. A monitor in $[a, b]$ changes by at most $(b - a)\delta_T$. An incumbent-deviation margin loses at most the sum of their two profile-specific payoff error bounds. Exact finite-prefix agreement preserves ever-downside and bounded discounted payoffs for a coherent countable-horizon execution. Summable mismatch bounds give ever-downside error at most $\min\{1, \sum_n \epsilon_n\}$.

Value-directed degradation can be sharper. Let $K_n, \bar{K}_n$ be original and comparison kernels, ψ_n a state projection, and $\bar{V}_n$ the comparison continuation value of a terminal monitor satisfying $f = \bar{f} \circ \psi_T$. With original state Z_n , define

$$r_n(z) = \bar{V}_n(\psi_n z) - K_n(\bar{V}_{n+1} \circ \psi_{n+1})(z).$$

The exact change is the initial projected-value discrepancy plus $\sum_n \mathbb{E}_{\text{original}} r_n(Z_n)$. Telescoping proves this identity. Each residual is bounded by projected-row total variation times the range of the continuation value. Changed observations with identical continuation values incur no loss; source-level margins require this calculation at both the incumbent and actual deviation profiles.

5.2. Payoff and temporal information can miss protection

Two physical laws in the running game are

$$\begin{aligned}\mu_{\text{safe}} &= \frac{1}{2}[\text{HH}] + \frac{1}{2}[\text{OO}], \\ \mu_{\text{risky}} &= \frac{7}{12}[\text{HH}] + \frac{1}{4}[\text{HO}] + \frac{1}{6}[\text{OH}].\end{aligned}$$

Both earn $(5/2, 2)$, but joint downside is respectively zero and $5/12$. A finite source-game embedding strengthens the distinction: let a fair hidden bit select which of two active commands is safe, with the other command risky and a third command opting out. Five role-1 checked sources select the safe command, the opposite command, either fixed command, or opt-out; their missing-receipt branches are committed. Role 2 has one checked mediator-following peer. Add the three constant physical-action sources to each role; checked sources opt out against unrecognized constants. Erasing the hidden bit preserves the entire 8×4 expected bimatrix and all mixed equilibria, but every erased equilibrium with role-1 payoff at least $5/2$ has joint downside $5/24$. Fine information admits a zero-downside equilibrium at that payoff. A finite proof is given in Appendix A.5.

Temporal marginals likewise leave lifetime protection undetermined. For $T \geq 3$, binary bad-round indicators can have $\mathbb{P}(B_t = 1) = 1/T$ and every pair independent, yet attain either sharp endpoint

$$\frac{T+1}{2T} \leq \mathbb{P}(\exists t : B_t = 1) \leq 1 - \frac{T-1}{T^2}. \quad (9)$$

At $T = 100$, the endpoints are 50.50% and 99.01%, compared with 63.40% for independent rounds. All have one expected bad round and matching pairwise laws. Lifetime guarantees therefore require the complete persistent resource process, not only one- or two-round measurements.

6. Experiments with model-written programs

6.1. Population design and measurements

The main experiments use Qwen3-8B in BF16 on H100 hardware, with native reasoning enabled, temperature 0.6, top- p 0.95, and output/context limits of 16,384/24,576 tokens. A Qwen3-14B follow-up repeats the stage-incentive intervention. Both checkpoints are pinned.¹ The model writes a bounded source before commitment; a finite expression interpreter acts during the game. Sources include readiness and physical-action expressions and optional inert notes. Arbitrary execution, imports, loops, and function calls are excluded. Every source pair’s complete quote/delivery law is evaluated with exact rational probabilities, and each tested deviation is authentically requested.

Each paired study has six population seed blocks and three writer positions per role. Arms share initial realized cohorts except in the prepared-initialization intervention. Revisions are synchronous: candidates are collected before a generation is committed. An invalid revision retains its incumbent; invalid initial attempts use a fixed fallback. Attempts are not repaired or resampled until valid. The first-wave studies have ten generations; follow-ups use four or six. Revision contexts contain committed sources, bounded final-answer memory, and exact own-payoff feedback when assigned. Native reasoning traces are not fed back. Fitness selection resamples within role after mutation, using exponential own-payoff weights and retaining ancestry.

High cooperation is the probability of HH in current-learner cross-play; low coordination is LL. Finite deviation gain is the maximum improvement from a registered source pool against assigned targets. It is a lower bound on unrestricted profitable deviation, and zero does not certify

¹8B revision b968826d9c46dd6066d109eabc6255188de91218; 14B revision 40c069824f4251a91eefaf281ebe4c544efd3e18.

unrestricted equilibrium. When targets are fixed, this gain need not describe incentives against current learners. *Worst sealed downside* is the maximum risk across every writer and every opponent in a twelve-source family withheld from feedback, averaged across seed blocks. A value of one means at least one certainly exposed writer–opponent pair per population, not certain harm for every writer. Source diversity counts canonical strings, which may share behavior.

The independent unit is the population seed. Reused cohorts are not pooled as independent replications, and within-population cross-play pairs and environmental atoms are not treated as independent samples. The results below summarize thirteen main studies and one separate delegation pilot; Table 1 highlights the main incentive controls.

Comparison	Generations	High cooperation	Low coordination	Deviation gain
Neutral resampling	6	0.111	0.426	0.694
Fitness resampling	6	0.000	0.944	0.056
Original payoffs, 8B	4	0.148	0.481	0.667
Reduced temptation, 8B	4	0.708	0.000	0.456
Original payoffs, 14B	4	0.046	0.537	0.442
Reduced temptation, 14B	4	0.338	0.208	0.597

Table 1: Endpoint means over six paired populations per comparison. Cooperation and coordination are probabilities; deviation gain is in payoff units against each arm’s assigned targets. Worst sealed downside is one in every population of every arm shown. Different studies and horizons are not pooled.

6.2. Selection favors low coordination, not replacement protection

At generation ten, the first selection study’s mean source diversity falls from 1 under equal weighting to 0.5278 under fitness selection, its primary endpoint; every paired seed decreases. Low coordination rises from 0.4491 to 0.7778, while finite deviation gain falls from 0.8333 to 0.2222. Three selected endpoint populations contain only literal L programs. These are exact current-population equilibria in the admitted physical-output language: against a literal L opponent, every source earns at most 2 pointwise, and literal L attains it regardless of quotation.

Every seed nevertheless has worst sealed downside one. A replacement opponent that always opts out makes a literal-L incumbent receive zero. The replacement receives one, less than its former coordination payoff two; the harmful replacement is not a profitable deviation. This separates equilibrium from replacement protection in observed populations.

A matched control resamples after mutation in both arms with shared random-number streams, using uniform weights in one arm and fitness weights in the other. After six generations, finite gain falls from 0.6944 to 0.0556 and low coordination rises from 0.4259 to 0.9444. Every seed has lower gain and greater low coordination under fitness weighting. High cooperation falls to zero; worst sealed downside remains one. Thus mechanical resampling alone does not account for the observed low-coordination selection effect.

6.3. Stage incentives increase cooperation with different stability effects

A paired intervention lowers exploiting payoffs $u_1(L, H) = 6$ and $u_2(H, L) = 5$ to 2, leaving high cooperation (4, 3) and the other outcomes unchanged. High cooperation becomes a strict stage equilibrium. The shared initial cohorts are retained, and the altered table is announced before revision.

After four generations at 8B, high cooperation increases from 14.8% to 70.8%, with four positive paired changes and two ties. Mean finite gain decreases, although one reduced-temptation endpoint still has gain 2.25. At 14B, cooperation increases from 4.6% to 33.8%, again in four

pairs with two ties, but finite gain increases from 0.4421 to 0.5972 and current-target downside rises from 0.1713 to 0.2083. Two 8B and one 14B treated populations consist entirely of literal H, certifying current-population equilibrium under the modified table. Worst sealed downside remains one throughout.

The intervention changes the strategic problem; it does not establish secure recognition in the original game. The checkpoints agree on the direction of cooperation but differ on stability and downside, rather than exhibiting a uniform scaling trend.

6.4. Fixed-source delivery effects differ from population comparisons

The joint-delivery study compares shared and independent delivery at equal marginals $r = 1/2$. Evolved endpoint downside is 0.1898 and 0.1817, respectively, with paired differences of both signs. This comparison changes sources as well as the channel.

Holding all twelve endpoint populations fixed instead gives 108 cross-role source pairs replayed under both couplings. One hundred have identical complete outcome laws; eight are sensitive. Equation 8 reproduces all 972 outcome-atom differences. Changing evaluation alone from shared to independent delivery increases mean downside by $1/432$ in shared-evolved cohorts and $5/432$ in independent-evolved cohorts, each concentrated in one seed. It reduces high cooperation by $1/216$ and $1/108$, respectively. Thus lower downside in the original between-population comparison coexists with higher downside in each frozen-source intervention. Most authored pairs ignore delivery coupling on their realized input domains.

6.5. Individual compliance improves while uniform protection fails

Both risk-objective arms receive exact risk feedback against two literal training sources, one always taking L and one always taking O. The constrained arm is instructed to maximize payoff subject to downside at most $1/20$ against *each* source, repairing violations first. Opt-out is feasible, but no external safety filter blocks a legal risk-violating revision. The primary endpoint is the generation-four maximum downside over all six writers and both trainers, compared within each seed.

That population maximum is one in every seed of both arms. No population has all six writers satisfying the cap. Secondary individual measures improve: average worst training risk falls from $29/36 = 0.8056$ to $19/36 = 0.5278$, improving in five populations and worsening in one. Compliant writer positions rise from $7/36 = 19.4\%$ to $17/36 = 47.2\%$. These positions remain clustered within the six population blocks.

Removing the exact training overlap leaves eleven disjoint sealed sources. A further evaluation uses 432 disjoint constant-action variants with altered syntax and notes. At each endpoint writer, worst risk is identical across training and both evaluation families. Thus the average improvement transfers to these finite probes, while uniform protection already fails on the training set. High cooperation rises from 0.0185 to 0.1111 as a secondary outcome; this does not repair the primary failure.

6.6. Further controls and a delegation pilot

A four-arm control crosses actual target dynamics, fixed or adaptive, with the writer’s announced dynamics. It uses common initial populations, payoff feedback, and ten revisions. The primary extended finite deviation search includes public sources, current learners, and adversarial sources, with every candidate requoted. Adaptation changes mean extended gain by -0.0579 under a fixed announcement and -0.0694 under an adaptive announcement; their interaction is -0.0116 . All three contrasts have both signs across seeds, with interaction ranging from -1.5 to 1.8333 . These outcomes show heterogeneous effects without establishing a consistent adaptation benefit or equivalence. Assigned targets differ across arms, so these are comparisons between strategic problems, not a common frozen evaluation.

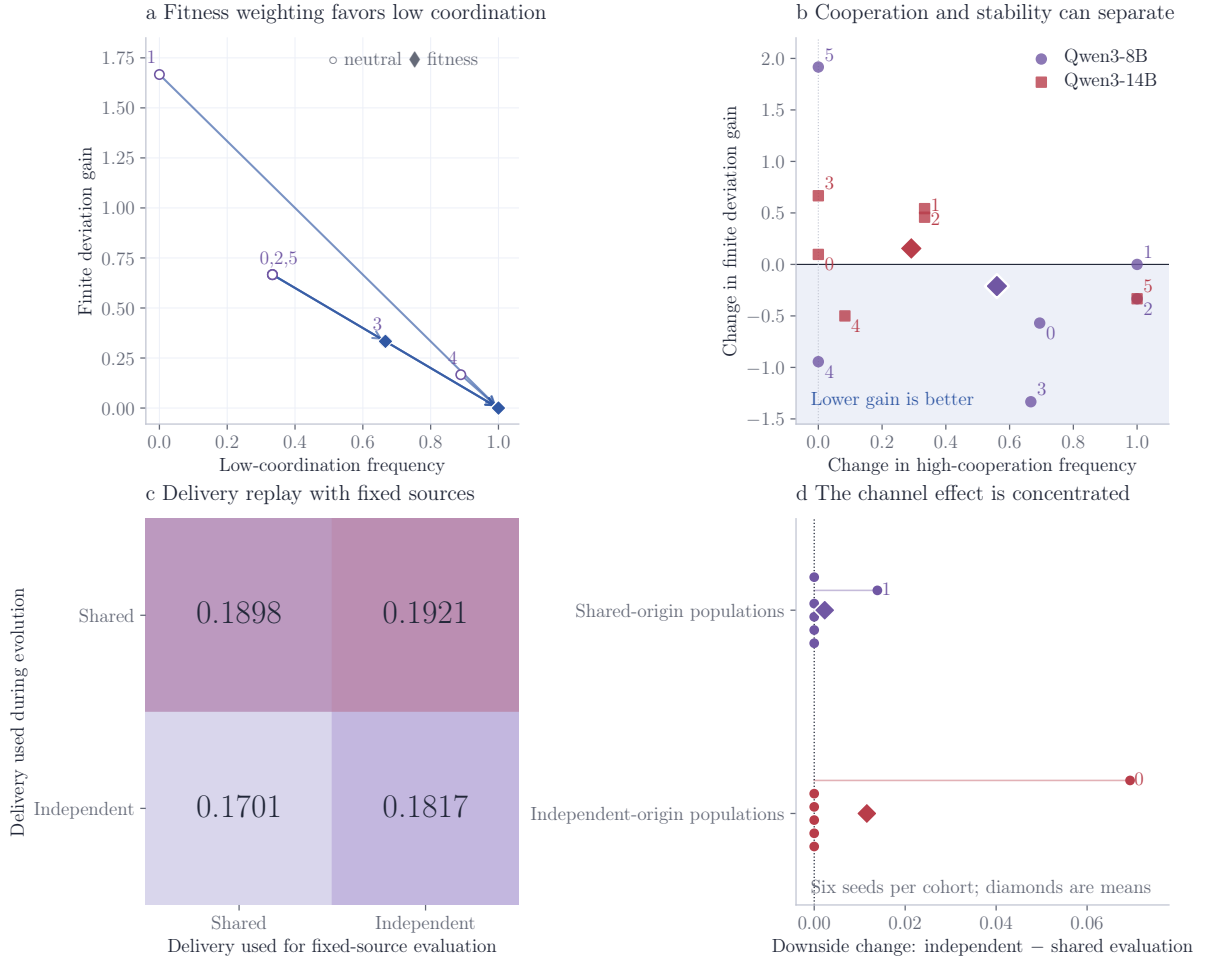


Figure 3: Three controls distinguish incentive and channel effects. **a**, neutral versus fitness re-sampling reduces tested deviation gain and increases low coordination in every pair. **b**, reduced temptation raises cooperation at both checkpoints, but residual deviation gain need not fall. **c–d**, frozen-source delivery replay separates the evaluation channel from the channel under which populations evolved; downside effects are concentrated in one seed per cohort. Diamonds show means and numbered points identify seed blocks.

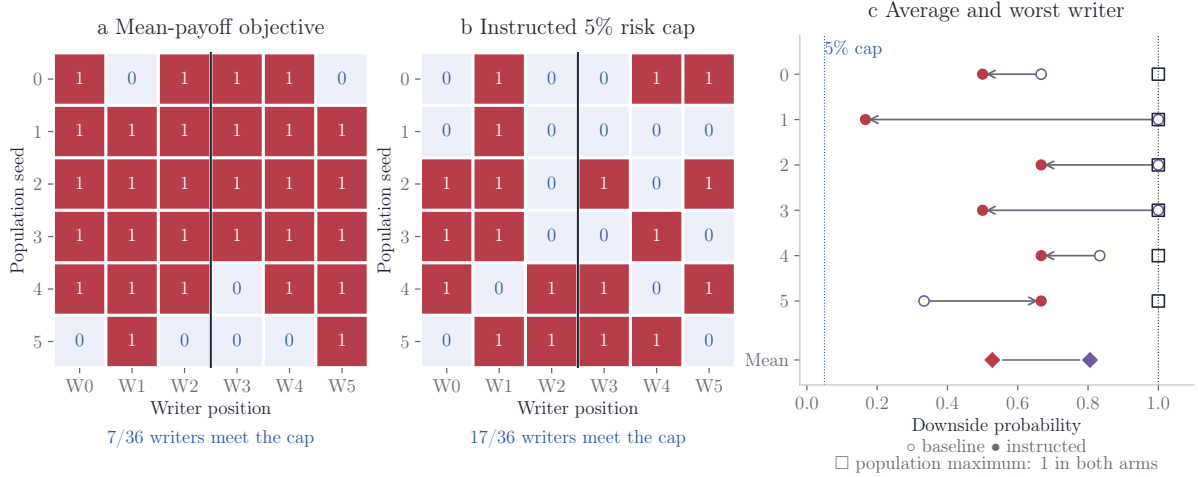


Figure 4: Risk-cap instruction at generation four. **a–b**, rows are population seeds and columns are writer positions; dark cells have worst training risk one, pale cells zero. Every row retains an exposed writer in both arms. **c**, individual-risk averages improve in five paired populations, but all population maxima remain one. Positions W0–W2 have role 1 and W3–W5 role 2.

Other first-wave interventions vary quotation, readiness reliability, payoff feedback, outsider pressure, and initial cooperation. High versus low quotation visibility increases mean endpoint cooperation by 0.2407, with a negative pair and a tie. Above- versus below-threshold readiness changes it by -0.0200 . Retry without payoff feedback has finite gain 0.4444 lower than the feedback arm. Greater information or feedback therefore does not uniformly produce better authored strategies; a threshold for a designated checked gate does not specify which source the model will select.

The separate delegation pilot gives cooperative play an authorization-checking stage with budgets $B = 0.2, 0.6, 0.8$ or exact authority. In its finite source language, role-2 best-response gains are respectively 0.276, 0.020, 0, 0, while role 1 has zero gain throughout. Nevertheless, all 192 model-written player outputs implement the same checked cooperative behavior, with zero primary behavioral contrast in all six blocks. All 24 guard-writing outputs fail the prescribed parser. The pilot demonstrates an incentive boundary without a corresponding population response; syntactic player validity alone does not establish best-response competence.

7. Related work and discussion

Blackwell’s comparison of experiments and channel-zonotope geometry compare information across decision problems [1, 2]. Our guarantee region restricts the supporting directions to nonnegative weights on specified casewise monitors. Functional compression uses characteristic graphs and hypergraphs to group compatible inputs [3, 4]; common-optimum obstructions also appear in Bayesian decision regret [5]. Here the one-interface budget is retained alphabet size, with explicit robust decoder constraints and output-quota exceptions.

State and history abstractions distinguish model, value, and policy preservation [6, 12, 13]. Value equivalence discards distinctions irrelevant to declared evaluation families [14, 15]. The adaptive criterion retains the actual owned-history tree and permits a compressed policy to dominate a fine policy rather than reproduce each named policy’s value. Robust cells use standard rectangular uncertainty and convex feasibility [7], while directed degradation follows value-aware simulation [16].

Program equilibrium already permits cooperation beyond stage equilibria [8, 9]. Partial trans-

parency and similarity signals can support cooperative equilibria and learned cooperation [10], and LLM-generated programs have been studied in open-source games and evolutionary populations [11]. Our source comparisons require authentic deviating quotes and actual joint resource laws, and protect downside monitors separately from expected incentives.

Three limitations delimit the claims. The exact policy-region criterion requires a finite, seed-blind, perfectly recalled control interface admitting all legal causal policies; an arbitrary autonomous source catalog need not have this closure. The repeated construction uses a sound bounded-language certificate and action-independent certified schedules; feedback can invalidate its common tail. The model experiments use small paired populations and finite adversarial families. They characterize authored behavior and frozen-source channel effects, but do not evaluate whether model-written programs learn the guarantee-preserving quotient or satisfy the repeated construction’s hypotheses.

The resulting distinction is operational: source identity is dispensable when the retained evidence supports common useful continuations, but truthful cooperative labels, matched mean payoffs, and matched temporal marginals do not by themselves establish that condition. Cooperative performance, strategic stability, and replacement protection require separate measurements and separate guarantees.

A. Proofs

A.1. Exact and oriented compression

The score of a table is $\sum_{e,a} T(e,a)b_e(a)$, maximized independently at each receipt. After garbling the score vector at z is $\sum_e \Gamma(z | e)b_e$. Subtracting the two optima and using row normalization gives equation (2). The output term is

$$\min_a \sum_e \Gamma(z | e) [\max_{a'} b_e(a') - b_e(a)],$$

which vanishes exactly when one action makes every positive-support difference zero. A deterministic lossless partition therefore gives precisely a proper conflict coloring. For a lossless stochastic map, choose one positive output per receipt; its new cells are subsets of the old compatible supports, so no extra labels are required. For any incompatible set and witnessing objective, select one receipt excluding each of the k actions. This supplies an incompatible subset of size at most k , establishing the minimal-edge bound.

For a zonotope, the fine support is $w^\top b + \sum_e [w^\top v_e]_+$. Let P_z, N_z be the positive and negative weighted masses at output z . The retained support is $w^\top b + \sum_z [P_z - N_z]_+$; the difference is $\sum_z \min(P_z, N_z)$, proving equation (3). Vanishing loss in every nonnegative direction is equivalent to equality of the compact downward regions. To see the converse directly, choose a fine guarantee x outside the retained region and its nearest retained point y . Downward closure gives $y \leq x$. With $w = x - y \geq 0$, the nearest-point segment derivative gives $w^\top (z - y) \leq 0$ for every retained z , while $w^\top x > w^\top y$. This supplies a distinguishing nonnegative support. In the binary case a conflict is witnessed by one positive and one negative generator, giving the graph criterion.

A.2. Robust cells

For fixed competitor a , write $\tilde{b}_e = b_e + d$. Its advantage over q is $b_e(a) - b_e^\top q + d^\top (e_a - q)$. Maximizing over the rectangle gives $rs_e \|e_a - q\|_1 = 2rs_e(1 - q_a)$, because q is a probability vector. Maximization over the finitely many competitors proves the regret formula and equation (4).

The cells are convex in the $(k - 1)$ -dimensional simplex. Suppose a minimal infeasible family has $n > k$ members; for each member i , choose x_i in all other members. Affine dependence gives coefficients λ_i , not all zero, with $\sum_i \lambda_i = 0$ and $\sum_i \lambda_i x_i = 0$. Normalized positive and negative coefficients give two convex representations of one point. For any member j , at least

one representation excludes x_j , hence lies entirely in that member. The common point belongs to every member, a contradiction. This proves the edge bound, including possible singleton infeasibility. Coloring and stochastic determinization follow as above. Decreasing r relaxes every inequality since $1 - q_a \geq 0$.

For the worked scores, $q = (1/2, 1/2, 0)$ has maximal regret $\max\{0.1 + r, -0.9 + 2r\}$ and is feasible through $r = 0.25$. Summing the first receipt's action-1 and second receipt's action-2 constraints gives $0.2 + 1.8q_3 + 2r(1 + q_3) \leq 0.7$, forcing $r \leq 0.25$. Either preferred deterministic action has worst regret $0.2 + 2r$ at the other receipt; the third fails nominally. The deterministic radius is therefore 0.075.

A.3. Adaptive preservation and its error budget

The finite performance set is the convex hull of deterministic causal tables: draw behavioral coins in advance for one inclusion. Conversely, a mixture of tables has an equivalent behavioral law under perfect recall. Conditional on a reached fine path, its environment likelihood given past commands is common across compatible tables because the environment cannot see the private table seed. This factor cancels. Remaining table weights depend only on remembered retained observations and commands. The same cancellation makes the conditional case/environment law at a fine history independent of the compatible policy reaching it.

Backward induction on that owned-history tree gives $V_w(h) = \max_a Q_w(h, a)$. At a reached history, subtract expected next optimal value from current optimal value and average under lifted retained occupancy. Summation over epochs cancels intermediate values, leaving the fine optimum minus terminal expected reward, proving equation (5). Every summand is nonnegative. A finite retained optimum exists, and its loss is zero exactly when it uses fine-optimal commands at each occupied history. Applying nonnegative support separation proves the region criterion. Domination of deterministic fine vectors implies domination of every mixture, and the converse follows from region equality. This proves Theorem 4.1.

For the approximate statement, domination $u \geq v - \varepsilon \mathbf{1}$ immediately gives $w^\top v \leq h_C(w) + \varepsilon \|w\|_1$. Conversely, separate a point outside the closed convex set $\mathcal{P}_C - \mathbb{R}_+^q + \varepsilon \mathbf{1}$. Downward closure again makes the separating normal nonnegative, contradicting the supporting bound. Positive homogeneity gives the least common error formula. Static receipt regret is summed with weights $\Gamma(z \mid e)$, whose rows sum to one. Adaptive regret is summed with actual occupancies, giving the stated accumulated bounds.

For Proposition 4.2, couple the original policy and replacement on the same case, resource, and private seed. Their identical pre-certificate histories give identical prefixes and monitor states. Conditional on each certified prefix, the label-specific tail dominates the original continuation. Summing these conditional inequalities proves casewise domination; if no certificate arrives they remain identical. Retained policies lift, proving the reverse region inclusion.

A.4. Repeated source guarantees and composition

Couple mode configurations at the same actual sources, opening time, and full joint ticket tape. Opaque receipts agree; disclosed identity parsing and the coarse sound bit give the same classification. Cooperative certified readiness depends only on this classification, clock, and ticket. Inductively their bits agree. Each fixed-mode outsider sees the same own ports, physical past, and private resources, so its bits and actions agree as well. Reliable readiness and the identical physical gate then give equality of the next physical histories, proving mixed-mode trace preservation.

Two gates produce HH on joint activation and OO otherwise, yielding their payoffs and lifetime safety. Against an outsider, disclosure produces permanent rejection. Exposure is therefore contained in initial blocking and at epoch n has probability at most $q_s(1 - r_s)^n$. Downside requires exposure. Against cooperative physical O, an outsider's best payoff is 1; against H

it is M_j . Thus $u_j \leq 1 + (M_j - 1)\mathbf{1}\{\text{exposure}\}$ pointwise, giving equation (6). A readiness-one/best-reply source attains its own actual-profile bound by taking L upon exposure and O otherwise.

Full cooperative readiness gives (4, 3) in every round. Cooperative source deviations remain on the HH/OO edge and cannot exceed the deviator's target payoff. Outside deviations obey their own source-specific ceiling. The inequalities in Theorem 4.3 exclude each profitable deviation; averaging over independently mixed counterpart sources proves the mixed statement.

For trace transport, couple the initial state and then each complete next row while pasts agree. Agreement survives with probability at least $\prod_n (1 - \epsilon_n)$; an event can differ only on mismatch. A bounded monitor differs by at most its range there. Incumbent-deviation margin bounds follow by subtracting two expectations. Ever-downside is an increasing union of finite-prefix events; bounded discounted sums have uniformly vanishing tails. Passing the finite bounds to these limits proves the countable-horizon claims. The directed identity follows by conditioning each residual on Z_n and telescoping; its row bound is the total-variation expectation inequality applied to the comparison continuation range.

A.5. Payoff and temporal separations

The two laws $\mu_{\text{safe}}, \mu_{\text{risky}}$ have identical payoff means by direct substitution in equation (1). Against the mediator-following peer, every active checked source therefore earns (5/2, 2) before and after hidden-bit erasure; the stopped source earns (1, 1). Checked-constant, constant-checked, and constant-constant profiles are unaffected. These exhaust the finite bimatrix, so all mixed equilibrium comparisons agree. The fine safe-command source and checked peer form a zero-downside equilibrium: active deviations tie, opt-out earns 1, and unrecognized literal L or H earns 0.

In either role, literal H is strictly dominated by a mixture assigning 2/3 to literal L and 1/3 to literal O, checked against all constant and checked counterparts. It cannot occur in equilibrium. Among remaining columns, literal role-1 L earns at most 2, and stopped or literal O earns 1. An active checked row earns $1 + (3/2)x$, where x is column mass on the checked peer. Thus role-1 payoff at least 5/2 forces $x = 1$ and all its row mass on active checked sources. Without the hidden bit, each active command is safe and risky equally often, giving downside $(1/2)(5/12) = 5/24$.

For equation (9), draw a count N and choose a uniform subset of N rounds to mark bad. The two count laws are

$$\begin{array}{c|ccc} & N=0 & N=1 & N=2 \\ \hline A & (T-1)/(2T) & 1/T & (T-1)/(2T) \end{array}, \quad \begin{array}{c|ccc} & N=0 & N=1 & N=2 \\ \hline B & (T-1)/T^2 & (T-1)/T & 1/T^2 \end{array}.$$

Both have $\mathbb{E}N = 1$ and $\mathbb{E}N(N-1) = (T-1)/T$, hence single-round probability $1/T$ and distinct-pair probability $1/T^2$. Their union probabilities are the two endpoints. For any process with these moments, the pointwise inequalities

$$\mathbf{1}\{N > 0\} \geq N - N(N-1)/2, \quad (N-1)(N-T) \leq T\mathbf{1}\{N=0\}$$

give the lower and upper bounds upon taking expectations. Both constructions attain them, proving sharpness without assuming independence beyond pairs.

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